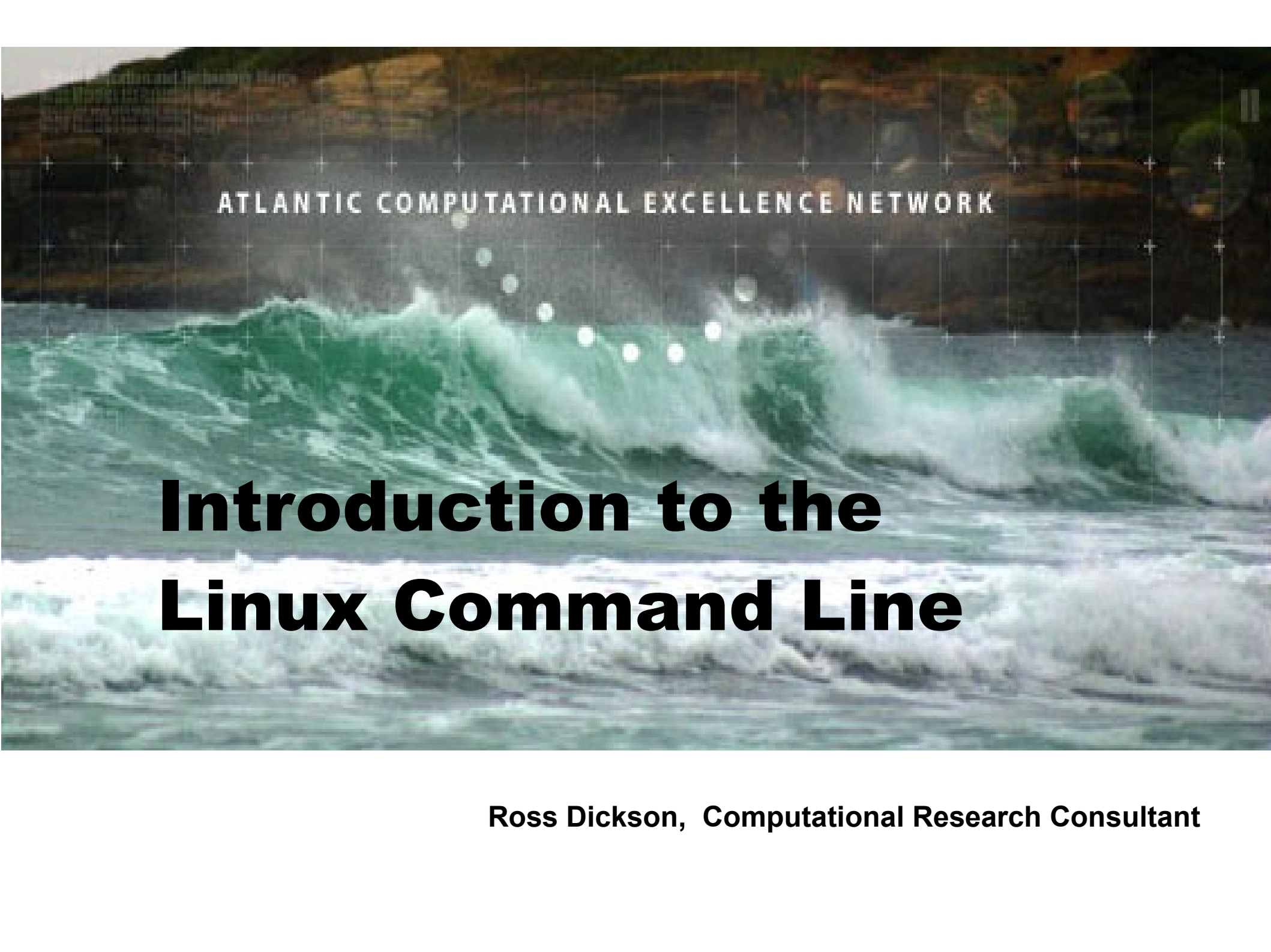




ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

Introduction to the Linux Command Line

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Outline

- Concepts
- Commands
- Files and directories
- Running programs

...not necessarily in that order

Login

Generally,

```
> ssh userid@host.ace-net.ca
```

Specifically, today:

```
> ssh userid@mahone.ace-net.ca
```

 mahone.ace-net.ca - PuTTY

login as: rdickson

rdickson@mahone.ace-net.ca's password:

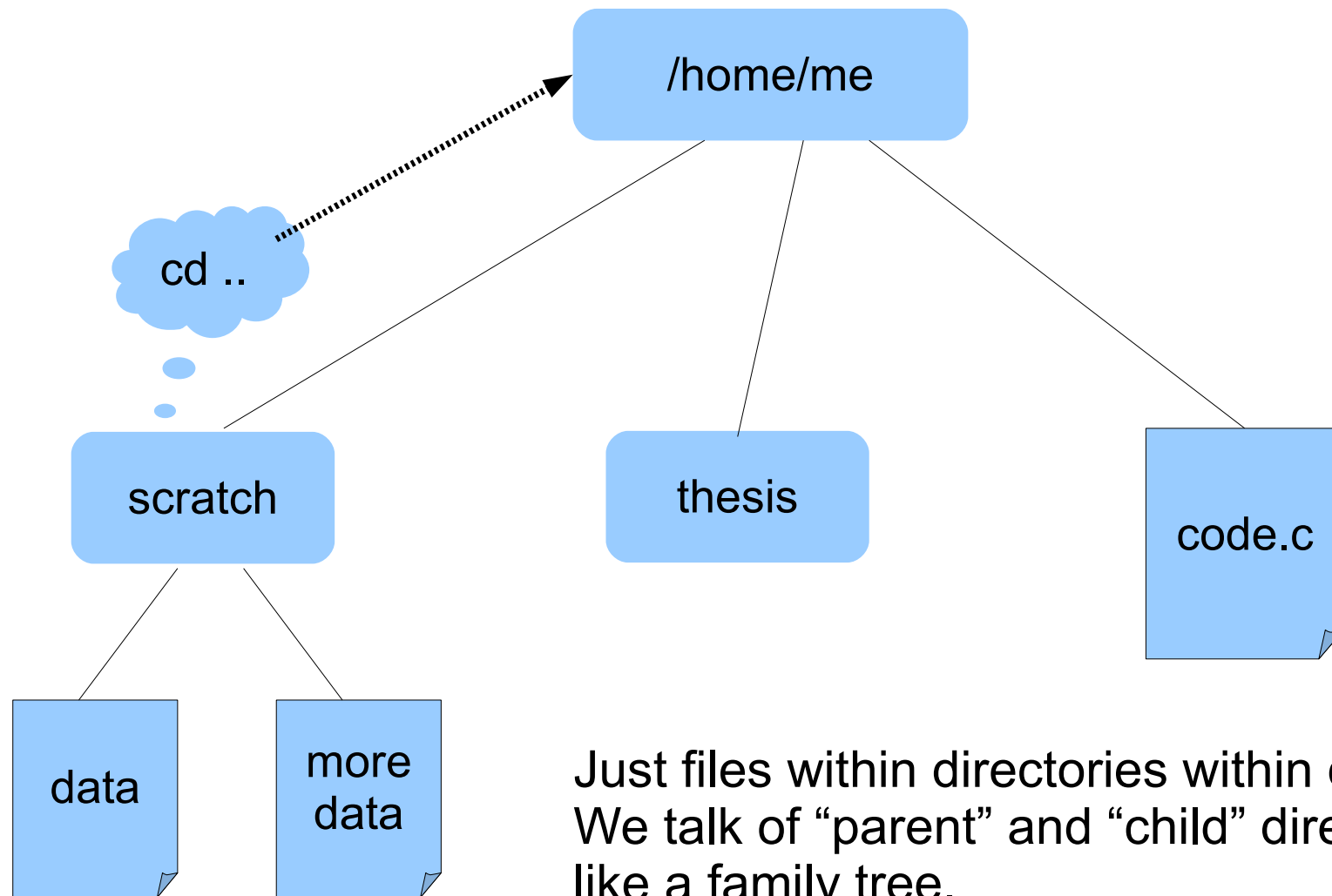
Last login: Fri May 7 10:33:45 2010 from rdickson.ucis.dal.ca

rdickson@mahone: ~ \$ 

Looking Around

- pwd
 - “print working directory”
 - where am I?
- ls
 - “list”
 - what files are here? what subdirectories?

Linux directories = Windows folders



Just files within directories within directories. We talk of “parent” and “child” directories, like a family tree.

Getting around

- `mkdir newdir`
 - “make directory”
- `cd directory`
 - “change directory”
- `cd ..`
 - .. means “parent directory” or “one level up”
 - ~ means “my home directory”, */home/userid/*
 - . means “right here”
 - / is separator, not \

Create your (empty) directory

```
> mkdir workshop  
> cd workshop  
> ls  
> pwd  
/home/userid/workshop  
>
```

Moving stuff around

- `mv from to`
 - “move”, also “rename”
- `cp from to`
 - “copy”
- `rm file`
 - “remove”
 - Careful --- There's no recycle bin!
- `rmdir directory`

Paths, directories, abbreviations

- directory separator is “/”
- /home/jsmith/code/test is an *absolute path*
 - begins with “/”
- code/test is a *relative path*
 - no “/” at the beginning
 - same thing if pwd is /home/jsmith
- *
- “wildcard”

Get some files

```
> pwd
```

```
/home/userid/workshop
```

```
> cp /home/rdickson/demo/C/* .
```

```
> ls
```

```
> ls
```

```
function.c  hello.c  input  int1.c
```

```
int2.c      int3.c  int4.c  Makefile
```

```
>
```

What's inside?

- `cat file`
 - just prints it all
- `more file`
 - use spacebar to page down
 - scroll through file, one screen at a time
 - q to quit, ctrl-B to back up

Editors

- **vi** and **vim**
 - **vi** often the default Unix editor
 - **vimtutor** is online tutorial for **vi** & **vim**
- **emacs**
 - powerful!
- **nedit**
 - requires X11 forwarding
- If you don't know which to choose, learn **vi**

vim

A screenshot of a PuTTY terminal window titled "mahone2.ace-net.ca - PuTTY". The terminal displays C code: "#include <stdio.h>", "main()", "{", " printf ("Hello World!\n");", "}", followed by several tilde (~) characters representing blank lines. The status bar at the bottom right shows "1, 1" and "All".

```
#include <stdio.h>

main()
{
    printf ("Hello World!\n");
}
~
~
~
~
~
~
~
~
~
~
~
~
```

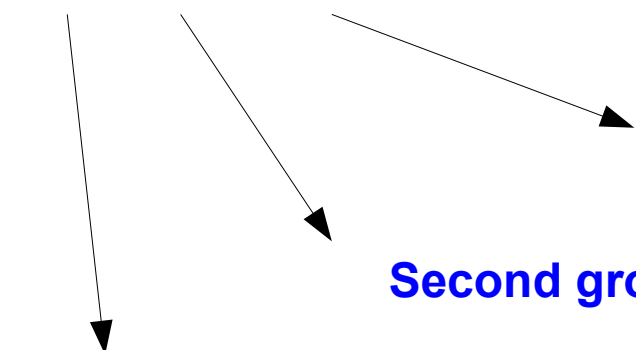
1, 1 All

File Permissions

Sometimes you need to edit the permissions of a file. The permissions of a file dictate who can execute, read and write to a file.

If you do **ls -l** you might see something like the following:

```
-rwr-xr-x  1 staveley TechTeam      271 Jul 12  2007 Makefile
```



Third group of 3: What everyone else can do

Second group of 3: What group members can do

First group of 3: what you (owner) can do --- read, write, execute?

- To see a directory listing you must have *execute* permission
- To change permissions, use **chmod**
 - **chmod +x file** to turn on execute permission for you
 - **chmod o-r file** to turn off read permission for others

Online manual

- `man command`
 - “manual”
- `man -k keyword`
 - searches online manual pages

What happens if I ...?

- which *program*
 - shows absolute path of *program*
- echo \$PATH
 - what directories are searched for executables?
 - \$PATH is an *environment variable*
- env
 - “environment”
 - shows all environment variables

Miscellany

More information can be found out about the system you are working on with the following commands:

du - disk usage - how much disk space is being used by the current directory

du -sh . - “summary, human readable, this directory”

df - how much disk space is available

date - displays the current date and time (machine clock)

qsub, qstat – submitting and monitoring jobs

Where to go for help

- Web tutorials
 - Google “Unix command line tutorial”
- Books
 - e.g. Glass & Ables, *Unix for Programmers and Users*, available at Dal bookstore
- ACEnet User Wiki
 - <http://wiki.ace-net.ca/>
- User support line
 - email: support@ace-net.ca